11 In this question you must show detailed reasoning.

The equation of a curve is $y = 5\sqrt{x} - x - 6$ for $x \geqslant 0$.

(a) Verify that the curve cuts the x-axis at $x = 4$ and at $x = 9$. **[2]**

The curve does not cut or touch the x-axis at any other points.

(b) Determine the exact area bounded by the curve and the x-axis. **[4]**